

NBA Home-Win Probability Model

Mathematics & Full Constant Provenance
Champion architecture `hfa_75` — technical briefing

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Prepared for recruiter / technical review.

Every number below is classified **Fixed**, **Selected**, **Fitted**, **Derived**, or **Numerical**, with an explicit source or calculation.

Artifacts: `artifacts/selected_spec.json`, `march_architecture_results.csv`, `coefficient_table.csv`.

Purpose of this document. This note states the model equations and accounts for **every numeric constant**: if a number is calculated, the exact optimization is written; if it is not calculated, the external or design source is named. No constant is left as an unexplained “magic number.”

1 Champion model (one equation)

$$p_g = \sigma\left(a \operatorname{logit}(p_g^E) + b \operatorname{logit}(p_g^R) + c\right)$$

with **deployed** stacker coefficients $a = 0.3592529607156558$, $b = 0.6407470392843442$, $c = 0.3253629989522476$, and $\sigma(z) = 1/(1 + e^{-z})$, $\operatorname{logit}(p) = \log(p/(1 - p))$.

These deploy coefficients satisfy $a + b = 1$ (temperature $\tau = 1$): the model is fitted by penalized maximum likelihood and then **projected onto a temperature floor** $\tau \geq 1$ so it never sharpens duplicate component information. The raw unconstrained MLE fit was $a = 0.5696035387238357$, $b = 1.0159186448349236$, $c = 0.3132108962839507$ ($\tau = 1/(a + b) \approx 0.63 < 1$); both sets are stored in `artifacts/selected_spec.json`. See §3.3 for the derivation and rationale.

Component probabilities p_g^E and p_g^R are themselves L2-logistic calibrations of Elo and (Bradley–Terry + trend) features built with architecture `hfa_75`.

2 How to read every constant

Class	Meaning
Fixed	Hardcoded design choice. Not estimated on this CSV.
Selected	Chosen by comparing candidates on March log loss .
Fitted	Numeric solution of an optimization / MLE on data.
Derived	Algebraic consequence of other constants (no new information).
Numerical	Solver / clip / RNG setting; not a modeling claim.

Honesty rule used below. If a number was not estimated or selected on this project’s data, this document says so. In particular, the MOV constant 2.2 is **Fixed** (literature), not **Fitted**.

3 Master ledger — every constant

3.1 A. Elo core (Fixed unless noted)

Constant	Value	Class	Provenance / calculation
Initial rating $R_i(0)$	1500	Fixed	Standard Elo origin. Code: <code>ratings = {team: 1500.0}</code> in <code>features.py</code> . Not fit.
Elo scale S	400	Fixed	Classical Elo logistic scale: a 400-point gap $\Rightarrow$ 10:1 odds in the raw Elo formula. Denominator in $x^E = (R_h - R_a + H)/400$.
Base of Elo logistic	10	Fixed	Classical Elo uses base-10 logistic, not e . Formula $1/(1 + 10^{-x})$.
MOV mode	<code>log</code>	Selected*	Architecture field <code>elo_mov</code> . All five candidates used <code>log</code> ; mode itself was not raced against <code>none</code> in selection.
MOV blowout shape	$\ln(m + 1)$	Fixed	FiveThirtyEight / Nate Silver NFL Elo log-MOV form.
MOV constant b_{MOV}	2.2	Fixed (external)	Adopted from the published FiveThirtyEight NFL Elo MOV formula (Silver). Not re-estimated on this CSV. Full professional rationale, benefits, and how to compute a data-optimal alternative: §5.4.
MOV slope a_{MOV}	0.001	Fixed	Paired with 2.2 in the same FTE formula. Ratio a/b sets favorite/underdog discount. Not fit here.
MOV denom floor	0.25	Fixed	Code guard <code>max(0.25, denom)</code> against pathological denominators. Not estimated.

*Selected only in the weak sense that the winning architecture kept `log`; no candidate used `none`.

3.2 B. Architecture hyperparameters for champion `hfa_75`

These appear in `configs/architecture_candidates.json` and the winner in `artifacts/selected_spec.json`. **Selection rule:** minimize March sequential log loss (`nba_wp/selection.py`). April rows loaded during selection: 0.

Constant	Value	Class	Provenance / calculation
K (Elo step)	10	Selected	Member of <code>hfa_75</code> candidate. Chosen because that architecture had lowest March LL among 5 candidates (Table 7). Not a continuous MLE.
H (home Elo pts)	75	Selected	Same: <code>hfa_75</code> beat $H \in \{55, 65\}$ variants on March LL. Equal-team home edge $\Rightarrow 1/(1 + 10^{-75/400}) \approx 0.606$.
C_{BT}	0.15	Selected	<code>sklearn LogisticRegression(C=0.15)</code> for Bradley–Terry. Value from winning architecture.
Trend half-life $\tau_{1/2}$	45 days	Selected	EWMA weight $2^{-\Delta \text{days}/45}$. From <code>hfa_75</code> .
Trend short window n_S	10 games	Selected	Mean of last $\min(10, n_i)$ margins. From <code>hfa_75</code> .
Elo logistic C_E	100	Selected	Regularization for Elo calibrator. From <code>hfa_75</code> .
Rank logistic C_R	0.1	Selected	Regularization for BT+trend calibrator. From <code>hfa_75</code> .
Architecture name	<code>hfa_75</code>	Selected	Argmin of March LL over 5 named bundles (not independent continuous search over each coordinate).

3.3 C. Stacker coefficients a, b, c (Fitted, then floored)

Constant	Value	Class	How calculated
a (deploy)	0.3592529607156558	Derived	Floored coef on $\text{logit}(p^E)$: $a = w/\tau_{\text{floor}}$. Scores April.
b (deploy)	0.6407470392843442	Derived	Floored coef on $\text{logit}(p^R)$: $b = (1 - w)/\tau_{\text{floor}}$.
c (deploy)	0.3253629989522476	Fitted	Intercept refit after flooring, given a, b fixed.

Constant	Value	Class	How calculated
a (raw MLE)	0.5696035387238357	Fitted	Unconstrained MLE coef on $\text{logit}(p^E)$. Selection surface only.
b (raw MLE)	1.0159186448349236	Fitted	Unconstrained MLE coef on $\text{logit}(p^R)$.
c (raw MLE)	0.3132108962839507	Fitted	Unconstrained MLE intercept.
τ_{floor}	1.0	Fixed	Temperature floor: deploy blend may not sharpen ($a + b \leq 1$).
Stacker C	1.0	Fixed	<code>LogisticRegression(C=1.0)</code> in <code>fit_logit_stack</code> . Penalty strength not selected on March.

Step 1 — unconstrained MLE (no closed form). Let March games be $g = 1, \dots, 239$. After fitting base models on dates $< 2026-03-01$ and scoring March component probs p_g^E, p_g^R ,

$$\hat{a}, \hat{b}, \hat{c} = \arg \max_{a,b,c} \sum_{g \in \text{Mar}} \left[Y_g \log p_g + (1 - Y_g) \log(1 - p_g) \right] - \frac{1}{2}(a^2 + b^2),$$

$$p_g = \sigma(a \logit(p_g^E) + b \logit(p_g^R) + c).$$

Solved by `lbfgs` (`sklearn`), giving the raw MLE (0.5696, 1.0159, 0.3132).

Step 2 — temperature floor. The two component logits are near-duplicates ($\rho \approx 0.97$), so the unconstrained fit yields $a + b = 1.586 > 1$, i.e. temperature $\tau = 1/(a + b) \approx 0.63 < 1$. A temperature below one *sharpens* the blend and produced extreme (99%+) out-of-sample prices — a leakage/overfit red flag. The model is therefore deployed with a floor $\tau \geq 1$. Preserving the Elo weight $w = a/(a + b) = 0.3593$ and setting $a + b = 1$:

$$a \leftarrow \frac{w}{\tau_{\text{floor}}} = 0.3593, \quad b \leftarrow \frac{1 - w}{\tau_{\text{floor}}} = 0.6407,$$

then c is refit by one-dimensional MLE with a, b fixed, giving $c = 0.3254$. The floor is enforced in `fit_logit_stack` and pinned by `tests/test_stack_temperature_floor.py`. Architecture selection still ranks candidates on the *unconstrained* March log loss; only the deployed model carries the floor.

3.4 D. Base-component logistic coefficients (Fitted)

From `artifacts/coefficient_table.csv` (April-scoring fit; base models refit through March for final April prices). Features are standardized inside the pipeline; both standardized and raw-unit coefficients are stored.

Component	Feature	Std. coef	Raw-unit coef
Elo	$x^E = \text{elo_diff}$	0.92651570	2.71292059
Elo	intercept	0.24151143	—
Rank	$x^{\text{BT}} = \text{bt_logit}$	0.80373649	1.50990421
Rank	$x^{\text{trend}} = \text{trend_diff}$	0.14417900	0.02999535
Rank	intercept	0.23352102	—

How calculated. For training rows available at that stage,

$$\hat{\beta} = \arg \max_{\beta} \sum_i \left[Y_i \log \sigma(\beta_0 + \beta^\top \tilde{\mathbf{x}}_i) + (1 - Y_i) \log(1 - \dots) \right] - \frac{1}{2C} \|\beta\|_2^2,$$

with $\tilde{\mathbf{x}}$ standardized (`StandardScaler`), $C = C_E$ or C_R , solver `lbfgs`. Raw-unit coef = standardized coef / training scale.

3.5 E. Bradley–Terry / trend auxiliaries

Constant	Value	Class	Provenance
BT design coding	+1/ − 1	Fixed	Home column +1, away −1. Standard BT contrast.
BT team strengths q_i, α	(vector)	Fitted	L2-logistic MLE on all prior games before date d , $C = C_{\text{BT}}$. Refit when history grows. Feature $x^{\text{BT}} = \hat{\alpha} + \hat{q}_h - \hat{q}_a$.
EWMA base 1/2	0.5	Fixed	Half-life definition: weight $2^{-\Delta/\tau_{1/2}} = (1/2)^{\Delta/\tau_{1/2}}$.
Beta prior (α, β)	(4, 4)	Fixed	Comment in code: “eight games of neutral information.” Smoothed win rate $(W + 4)/(G + 8)$. Used for <code>record_logit</code> (ablation feature; not in champion X).
Rest cap	7 days	Fixed	Rest days clipped to $[0, 7]$. Schedule features only (ablation path).
B2B threshold	1 day	Fixed	<code>back_to_back = (rest <= 1)</code> . Ablation path.
Density windows	4, 6 days	Fixed	Games-in-window counts. Ablation path.

3.6 F. Calendar / selection policy (Fixed policy)

Constant	Value	Class	Provenance
Train-through (base, March scoring)	2026-02-28	Fixed	<code>selection_policy.json</code> / selection code.
Selection window	2026-03-01–03-31	Fixed	Same.
April holdout start	2026-04-01	Fixed	Contract: selection raises if April present.
Accuracy cutoff	0.5	Fixed	$p \geq 0.5$ in metrics. Not optimized.
Primary metric	log loss	Fixed	Selection rule text.
# architectures raced	5	Fixed	Length of candidate list.
		design	

3.7 G. Derived quantities (Derived)

Quantity	Value	Calculation
$w = a/(a + b)$ (raw)	0.3593	$0.56960354/(0.56960354 + 1.01591864)$; preserved by the floor
$\tau = 1/(a + b)$ (raw)	0.6307	$1/(0.56960354 + 1.01591864) < 1 \Rightarrow$ floor applied
τ (deploy)	1.0000	$1/(0.35925296 + 0.64074704)$ after projection onto $a + b = 1$
$s = c$ (deploy)	0.3254	Intercept refit after flooring
Equal-team home p	≈ 0.606	$1/(1 + 10^{-H/400})$ with $H = 75$
Fair home odds	$1/p_g$	Definition
Fair away odds	$1/(1 - p_g)$	Definition

3.8 H. Numerical / solver settings (Numerical)

Setting	Value	Where
<code>max_iter</code>	10000	all logistics
<code>tol</code>	10^{-12}	all logistics
<code>random_state</code>	0	reproducibility
<code>solver</code>	lbfgs	all logistics
probability clip (logit)	$[10^{-12}, 1 - 10^{-12}]$ or $[10^{-9}, 1 - 10^{-9}]$	<code>model.py/features.py</code>

4 Target and information

$$Y_g = \begin{cases} 1, & \text{home points}_g > \text{away points}_g, \\ 0, & \text{otherwise.} \end{cases} \quad X_g = f(\mathcal{H}_{<d}).$$

Same-date batching: write all rows for date d before any Elo/BT/trend update from results on d .

5 Elo mathematics

5.1 Raw probability and feature

$$p_g^{\text{Elo,raw}} = \frac{1}{1 + 10^{-(R_h - R_a + H)/400}}, \quad x_g^E = \frac{R_h - R_a + H}{400}.$$

Constants: $R_i(0) = 1500$ (Fixed), 400 (Fixed), $H = 75$ (Selected).

5.2 Update with MOV

$$R'_h = R_h + KM_g(Y_g - p_g^{\text{Elo,raw}}), \quad R'_a = R_a - KM_g(Y_g - p_g^{\text{Elo,raw}}),$$

$$M_g = \ln(|m_g| + 1) \cdot \frac{2.2}{\max(0.25, 0.001(R_{\text{win}} - R_{\text{lose}}) + 2.2)},$$

where $R_{\text{win}}, R_{\text{lose}}$ are the pregame ratings of the *winning* and *losing* team (§5.3). Constants: $K = 10$ (Selected); 2.2, 0.001, 0.25 (Fixed).

5.3 MOV rating-difference sign (winner – loser)

The rating difference inside the MOV denominator is the *winner minus loser* rating, not a fixed *home minus away* rating. This matches the published FTE form ($\text{ELO}_W - \text{ELO}_L$) and has two correct consequences: (i) an **upset** (a lower-rated team winning) gives a smaller denominator and hence a *larger* rating update than an expected result, and (ii) with zero home advantage the update is exactly team-swap symmetric. Using a fixed home–away difference instead would make the autocorrelation correction depend on venue rather than on who actually won, under-updating home upsets and over-updating home favorites. This behaviour is pinned by `tests/test_elo_mov_winner_diff.py`.

5.4 The constant 2.2: source, case for use, and how to optimize it

Disclosure (external source). The value 2.2 was **not estimated from this project’s NBA dataset**. It was adopted from a published, widely cited sports-rating specification: Nate Silver / FiveThirtyEight, *Introducing NFL Elo Ratings* (archived: <https://web.archive.org/web/20171004105151/https://fivethirtyeight.com/features/introducing-nfl-elo-ratings/>), which defines

$$M = \ln(|PD| + 1) \cdot \frac{2.2}{(\text{ELO}_W - \text{ELO}_L) \cdot 0.001 + 2.2}.$$

This submission copies that functional form into `nba_wp/features.py::elo_multiplier` when `elo_mov="log"`. Transparency on this point is intentional: industry models routinely reuse validated structural constants when the *form* is what matters and the local sample is too short to re-estimate every shape parameter cleanly.

Professional case for using 2.2 here (best case, optimal or not). Even though 2.2 is external, using this specific published constant is a defensible engineering choice for a leakage-audited pricing model:

1. **Correct economic shape, not an arbitrary fudge.** The log term gives diminishing returns to blowouts (a 30-point win should not update ratings like three independent 10-point wins). The fraction involving 2.2 implements the standard underdog/favorite correction so favorites that “run up the score” do not inflate permanently. That is precisely the behavior a book wants in a strength tracker feeding probabilities.

2. **Stability and reproducibility.** A public constant with a fixed formula is auditable: two reviewers regenerating features get the same Elo path. Replacing it with an in-sample tuned value on one partial season would add a free parameter, widen the selection surface, and invite overfit to March noise.
3. **Separation of concerns in the validation design.** This project deliberately concentrates selection effort on quantities that move prices most (home advantage H , step size K , BT/trend regularization) under a March log-loss rule, while holding the MOV *shape* at a known industry default. That is a standard bias–variance trade: fix low-sensitivity shape parameters; select high-sensitivity ones.
4. **Empirical support that the choice is not fragile.** On this repository’s champion architecture, disabling MOV entirely worsens March log loss (≈ 0.4893 vs ≈ 0.4880 with the log-MOV form). Varying the constant in a neighborhood of 2.2 changes March LL only in the fourth decimal place (flat likelihood). So the *presence* of this MOV correction is beneficial; the exact published 2.2 is a stable operating point, not a knife-edge claim of global optimality.
5. **Interpretability for stakeholders.** 2.2 has a clear reading in the literature as the baseline expected $\ln(|PD|+1)$ scale in the autocorrelation correction (see Morse’s derivation of $d = 2200 = 2.2/0.001$). That is preferable to an opaque tuned scalar with no external meaning.

How to calculate the value that would have been most optimal. If the goal is a *data-optimal* MOV constant b^* for this pipeline (rather than the published 2.2), compute it by the same selection principle already used for architectures—minimize March sequential log loss—while holding the champion architecture fixed:

1. Fix architecture `hfa_75` except replace the literal 2.2 by a candidate $b > 0$ in

$$M_g(b) = \ln(|m_g| + 1) \cdot \frac{b}{\max(0.25, 0.001(R_h - R_a) + b)}.$$

(Optionally keep the literature ratio by setting the slope to $b/2200$ instead of 0.001; the one-parameter profile below is the minimal change.)

2. For each candidate b on a grid (e.g. $b \in \{1.0, 1.5, \dots, 4.0\}$) or by continuous search: build features $\rightarrow$ fit base models on games before 2026-03-01 $\rightarrow$ fit the March stacker $\rightarrow$ record March log loss $LL(b)$.
3. Set the optimal replacement

$$b^* = \arg \min_b LL_{\text{March}}(b)$$

That is: choose the MOV constant b that minimizes March sequential log loss. This b^* is the March-optimal MOV constant for this model and metric. On a quick profile of the current code path, $LL(b)$ was minimized toward the high end of the tested grid (near $b = 4$), with only a tiny improvement over 2.2 ($\Delta LL \sim 4 \times 10^{-5}$). Hence 2.2 is not claimed as b^* ; it is claimed as a published, stable default whose March performance is essentially tied with nearby optima.

4. **Stronger production estimator (multi-season).** On a longer NBA history, estimate the literature intercept directly: regress bin-wise $\widehat{\mathbb{E}}[\ln(|PD| + 1) \mid \text{win}, x]$ on Elo gap x and take the intercept $\hat{b}_0$ (and slope $\hat{a}_0$) as in the Morse/FTE construction, then set $b \leftarrow \hat{b}_0$. Alternatively, nest b inside the architecture search and select by prequential log loss with a locked final holdout.

One-sentence position for reviewers. *I used the published FiveThirtyEight log-MOV constant 2.2 as a fixed, auditable shape parameter because it encodes a validated blowout/underdog correction, keeps the Elo path reproducible, and sits on a flat region of the March likelihood; a March (or multi-season) profile of $LL(b)$ is the correct way to replace it with a data-optimal b^* if that becomes a production requirement.*

5.5 Calibrated Elo probability

$$p_g^E = \sigma(\beta_0^E + \beta_1^E \tilde{x}_g^E), \quad C_E = 100 \text{ (Selected).}$$

6 Bradley–Terry + trend

6.1 Bradley–Terry

Prior games before date d :

$$P(Y = 1) = \sigma(\alpha + q_h - q_a), \quad C_{\text{BT}} = 0.15 \text{ (Selected)}.$$

Feature: $x_g^{\text{BT}} = \hat{\alpha} + \hat{q}_h - \hat{q}_a$ (**Fitted** each day).

6.2 Trend

$$L_{i,d} = \frac{\sum_{k < d} 2^{-\Delta_{kd}/45} m_{ik}}{\sum_{k < d} 2^{-\Delta_{kd}/45}}, \quad S_{i,d} = \frac{1}{n_i^{\text{short}}} \sum_{\text{last } n_i^{\text{short}}} m_{ik},$$

$$n_i^{\text{short}} = \min(10, n_i), \quad T_{i,d} = S_{i,d} - L_{i,d}, \quad x_g^{\text{trend}} = T_{h,d} - T_{a,d}.$$

Constants 45 and 10: **Selected** with `hfa_75`. Base 2 in $2^{-\cdot}$: **Fixed** half-life algebra.

6.3 Rank component

$$p_g^R = \sigma(\beta_0^R + \beta_{\text{BT}} \tilde{x}_g^{\text{BT}} + \beta_{\text{tr}} \tilde{x}_g^{\text{trend}}), \quad C_R = 0.1 \text{ (Selected)}.$$

7 Selection proof for architecture constants

March sequential metrics from `artifacts/march_architecture_results.csv` (lower LL wins):

Architecture	K	H	March LL	Winner
hfa_75	10	75	0.48802866	yes
balanced	10	65	0.48805432	
lower_bt_shrinkage	10	65	0.48816185	
conservative	7.5	55	0.48871458	
responsive	15	65	0.49037595	

This is the **entire calculation** behind calling $K = 10$, $H = 75$, $C_{\text{BT}} = 0.15$, $\tau_{1/2} = 45$, $n_S = 10$, $C_E = 100$, $C_R = 0.1$ “selected”: they are the coordinate values of the bundle with minimal March LL. They are not independently profiled.

8 Metrics (definitions; cutoffs Fixed)

$$\text{LL} = -\frac{1}{N} \sum_g [Y_g \log p_g + (1 - Y_g) \log(1 - p_g)], \quad \text{BS} = \frac{1}{N} \sum_g (p_g - Y_g)^2.$$

Accuracy uses Fixed cutoff $p_g \geq 0.5$.

Champion results (artifacts). **Primary April holdout (frozen pre-April)**: LL 0.484432, Brier 0.155823, AUC 0.862798, accuracy 81.25% ($N = 96$). March selection surface (unconstrained stacker, $N = 239$): LL 0.48802866; the deployed (floored) stacker scores March LL 0.508373. Optional April sequential backtest (live-update simulation): LL 0.474545.

9 End-to-end constant flow

$$\underbrace{1500, 400, 2.2, 0.001}_{\text{Fixed Elo}} + \underbrace{K, H, C_{\text{BT}}, 45, 10, C_E, C_R}_{\text{Selected bundle}} \xrightarrow{\text{features}} (x^E, x^{\text{BT}}, x^{\text{trend}}) \xrightarrow[\text{Fitted}]{\beta' s} (p^E, p^R) \xrightarrow[\text{Fitted}]{a, b, c} p.$$

Code map. `nba_wp/features.py` (Fixed Elo/MOV/Beta; Selected architecture fields), `nba_wp/model.py` (Fitted β , a, b, c), `nba_wp/selection.py` (Selected bundle via March LL), `artifacts/selected_spec.json` (persisted winner + a, b, c).